

WASP-12b

R-0206

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_250104 · created 2026-07-16T23:00:24+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460679.8601 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.045
Transit depth (Rp/Rs)^2	2.36 +/- 1.38 [%]
Orbital Inclination (inc)	83.17 +/- 2.89
Airmass coefficient 1 (a1)	1.06 +/- 0.68
Airmass coefficient 2 (a2)	-0.05 +/- 0.48
Scatter in the residuals of the lightcurve fit is	63.66 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	3.0
Transit Duration (day)	0.1026 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.045 vs 0.1178 (deviation 0.58σ)
Duration fit vs published	0.1026 d vs 0.125 d (deviation 3.68σ)
Mid-time O–C	-2.9 min
Depth SNR	2.5
Residual scatter	63.66 %

Light curve

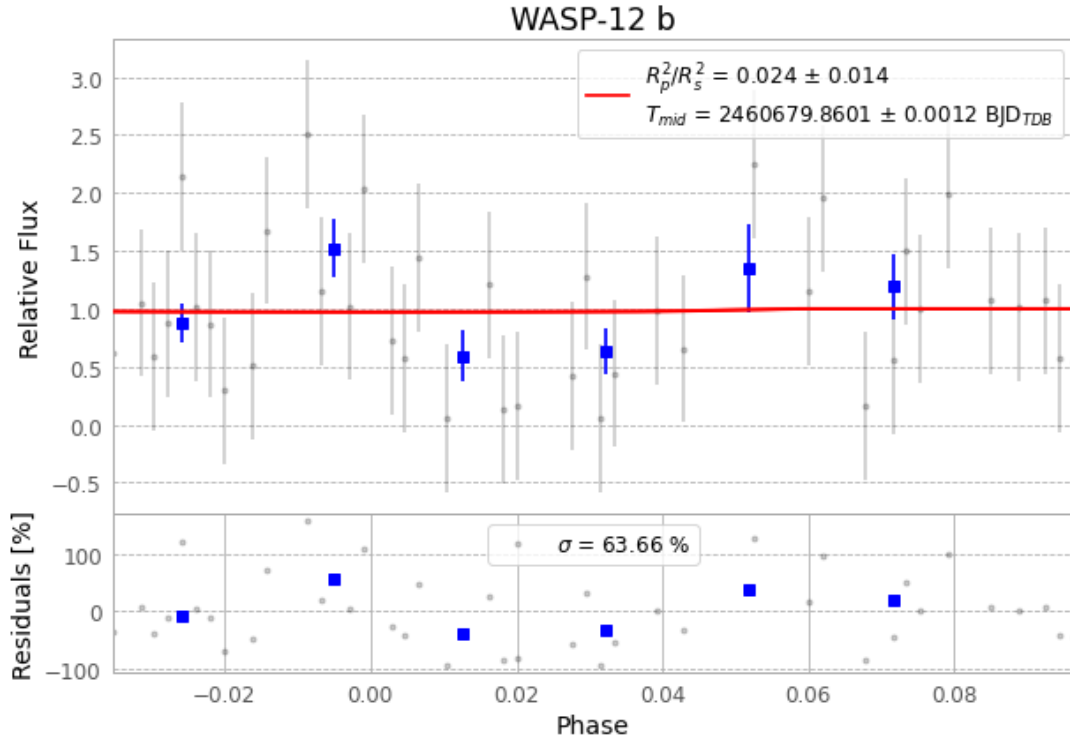


Figure 1 — FinalLightCurve_WASP-12 b_2025-01-04.png (SHA-256 7a5a77bbf2a274d9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [410, 328], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 951be76aa9830b75...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1965f5e

Data provenance

72 FITS frames (47 MB), WASP-12250104073316.FITS → WASP-12250104110615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-250104.log (016454a7f72a...), FinalLightCurve_WASP-12 b_2025-01-04.png (7a5a77bbf2a2...), FinalLightCurve_WASP-12 b_2025-01-04.csv (44c8966a5ed7...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 23:00Z.